Jairaj Narendran

Mechanical Design Engineer | Mechanisms, Prototyping & Testing

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WORK EXPERIENCE

Boston University **Boston, MA**

Visiting Researcher, Graduate Researcher *September 2021 – Present*

Grasping Delicate Objects with Kirigami Grippers – BU Ignition Award Startup Fund

- Owned end-to-end development of a robotic gripper for delicate objects in medicine and agriculture, delivering 8X longer fatigue life (80,000+ cycles) and 6X higher load capacity (300+ g)
- Designed, fabricated, and iterated 3D-printed prototypes using CAD (SolidWorks, Onshape), applying GD&T (ASME Y14.5) and tolerance stack-up analysis, maintained detailed drawings and specifications for traceability
- Drove failure investigations on load-bearing components to guide material and geometry selection, developed dynamic-loading test methods and led root-cause analysis of test failures for corrective design changes
- Discovered a snapping closure mode by adding origami folds to the kirigami gripper, enabling rapid-cutting uses such as produce harvesting; presented at NEW.Mech 2023

Bio-inspired Restoration of Aged Concrete Edifices (BRACE) – DARPA-funded program with Battelle, MIT, and SDSM&T

- Developed a new construction process that made fiber-confined structural systems for airfield damage repair 5X faster to build, collaborating cross-functionally with research engineers, subject-matter experts, and biochemists
- Led BU's technical workstream in a 5-institution, 15+ member program, as engineering point of contact, resolving critical production issues and reporting test data in monthly reports and biweekly presentations
- Reached 1,000 psi against the 400 psi ASTM C39 compressive-strength milestone (2.5X target) through structured root-cause analysis and iterative experimentation

Compliance-Shifting Elastogranular Shells – Thesis Work

- Designed, fabricated, and tested pneumatically actuated bilayer shell structures achieving 3.5X tunable stiffness and 2X buckling load, with a 36 g shell supporting a 500 g load
- Built MATLAB data-processing pipelines to evaluate structural integrity and mechanical performance under dynamic loading conditions, linking geometric and material parameters to structural behavior
- Implemented Python Monte Carlo simulations of grains packed on hemispherical shells to map how load transfers between grains, validated against physical testing to guide design decisions
- Mentored 25+ students and taught instrumentation, data acquisition, and controls to 80+ students (4.6/5 rating)

EDUCATION

Boston University **Boston, MA**

- Ph.D., Materials Science and Engineering *May 2026*
- M.S., Materials Science and Engineering *January 2026*

University of Illinois Urbana-Champaign **Champaign, IL**

- B.S., Bioengineering, Minor in Materials Science and Engineering *May 2021*

SKILLS

- **Mechanical Design:** CAD (SolidWorks, Onshape), GD&T (ASME Y14.5), tolerance stack-up analysis, DFM, 3D printing (FDM, SLS), machining, laser cutting, silicone molding, rapid prototyping, UR5 robotic arm
- **Engineering Analysis & Test:** failure mode & root-cause analysis, load path & structural analysis, statistical analysis, DOE, dynamic mechanical & fatigue testing, electronic test equipment, test protocol development, SOPs
- **Software:** MATLAB, Python, LabVIEW, Arduino, Adobe Illustrator, Microsoft Office 365 (PowerPoint, Word, Excel)